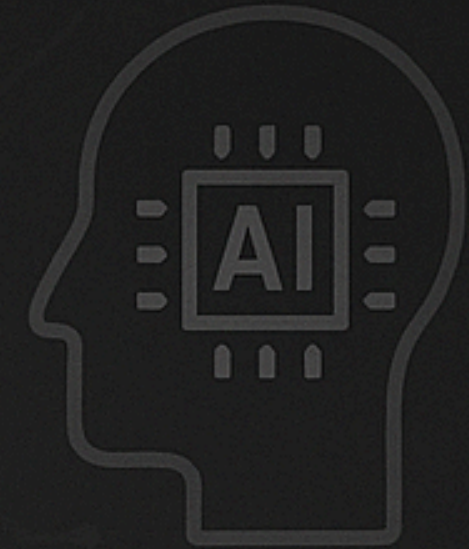


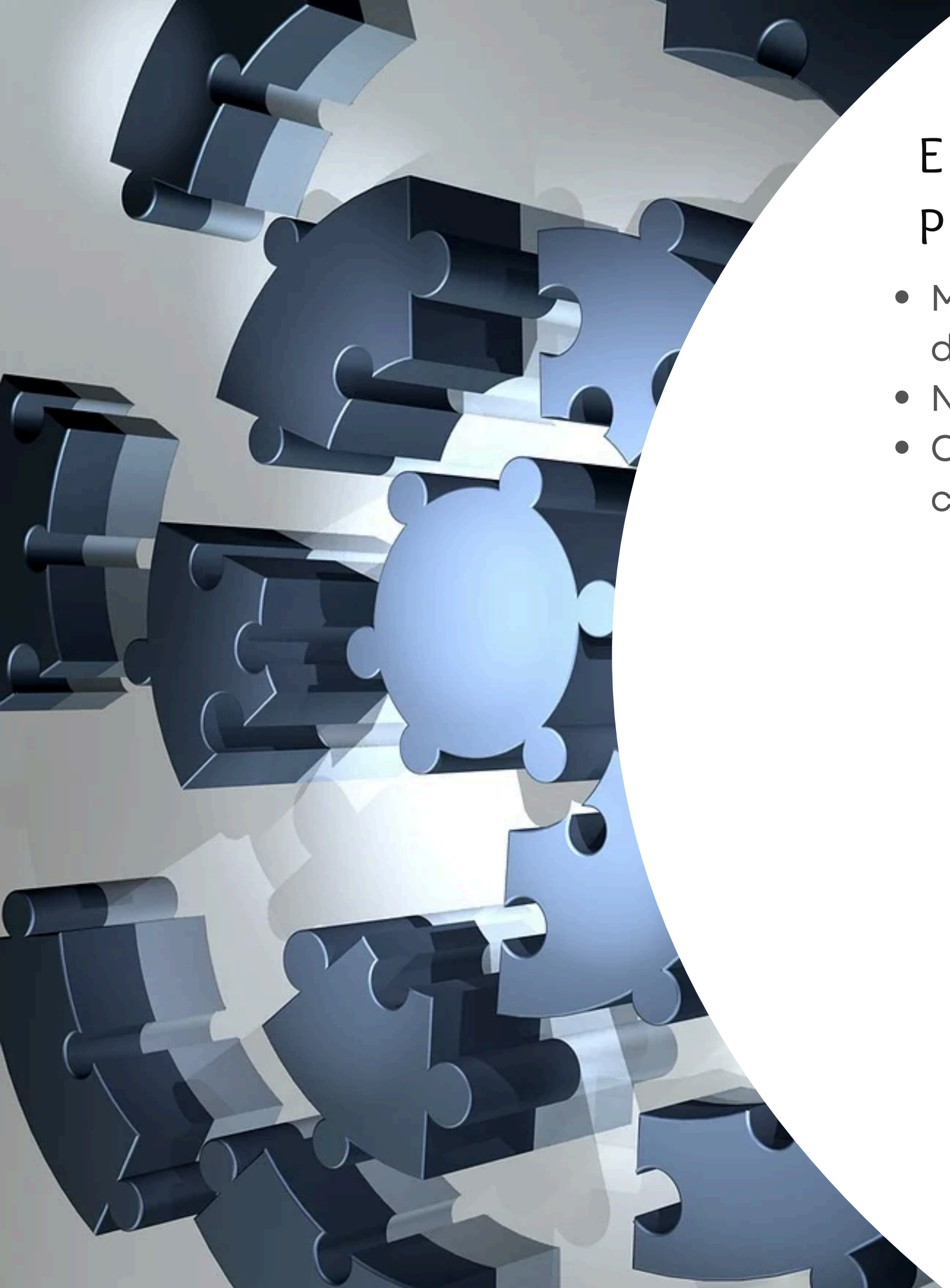


TIME
02:15

AMOUNT
₹ 5.75



PARKSMART



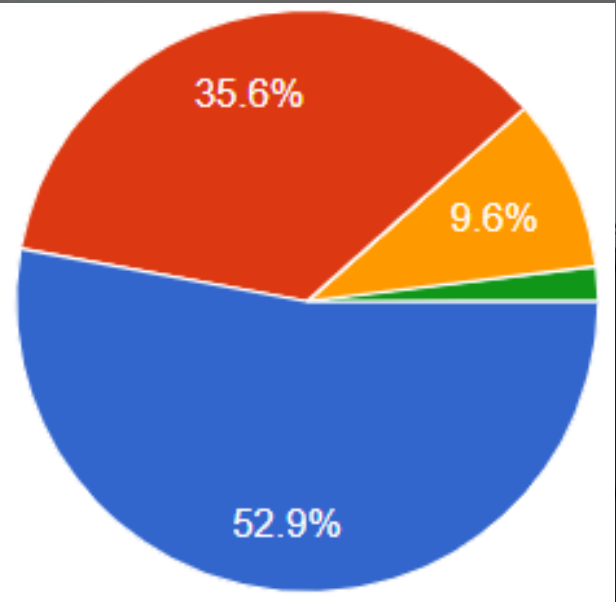
EVERYDAY PARKING PROBLEM

- Manual parking systems often cause confusion, delays, and unfair billing.
- No transparency in charges
- Cash payments and paper tickets are still common – inconvenient in the digital age.

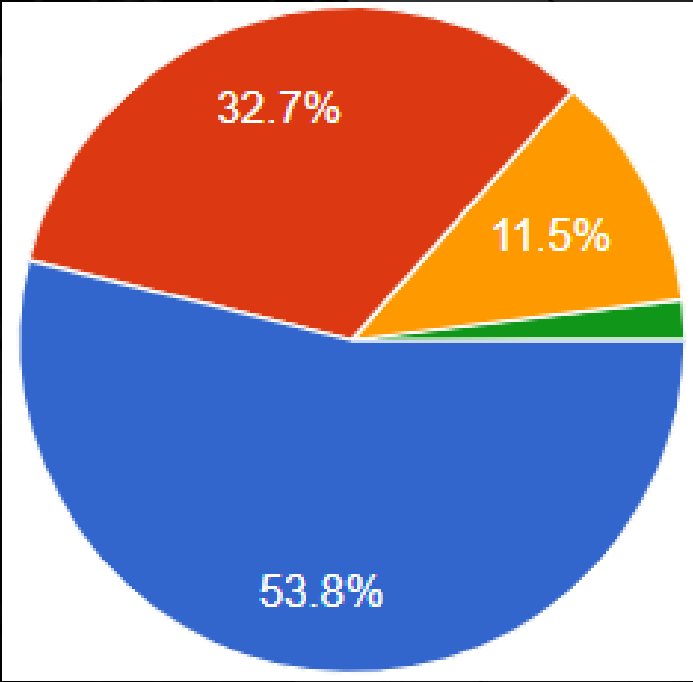
OUR SOLUTION: SMART AND SIMPLE

- Automatic entry & exit detection through number plate recognition.
- Instant digital billing – no human error.
- Real-time parking status.
- Online payments through UPI or wallets.
- Accessible on any device, anywhere, anytime.

People preferring a Full automated system

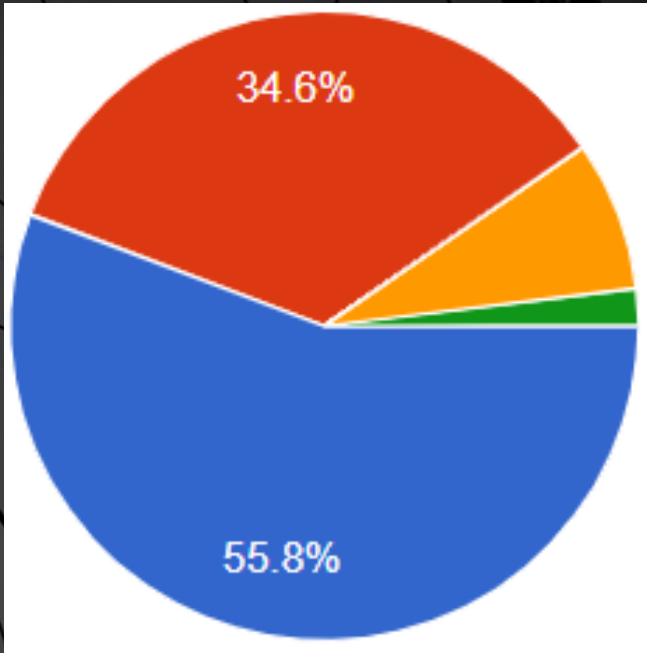


People wanting a fully transparent billing

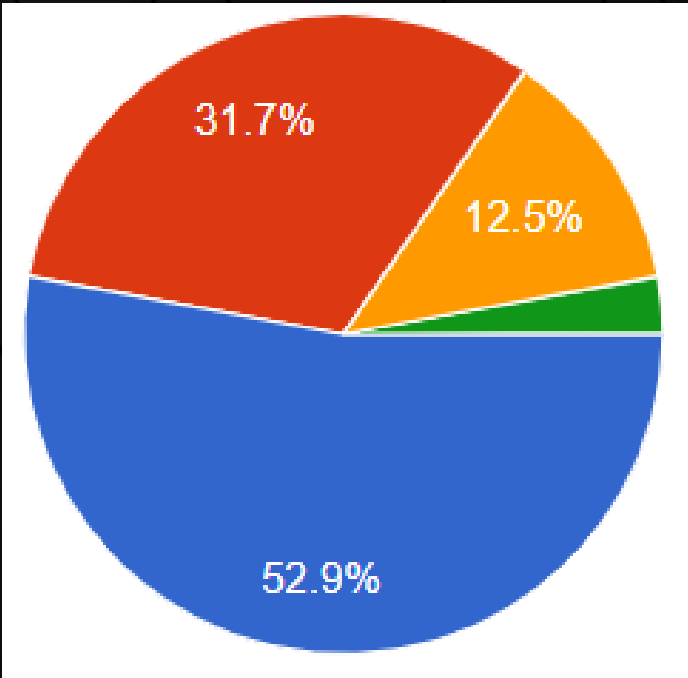


SURVEY RESULTS

People faced unfair billing due to slow cashier, congestion, want a bill to stop nearly when leaving the spot



People preferring a system they can access anywhere, anytime





INTRODUCING

PARK SMART

OVERVIEW

A web-based smart parking system that uses Automatic Number Plate Recognition (ANPR) to detect vehicles, track parking time, and enable online billing & payment.

TECH STACK

LAYER	TECH USED	FUNCTION
FRONTEND	HTML, TAILWIND CSS, JS	User UI for login, dashboard, billing
BACKEND	FLASK(PYTHON)	Logic, APIs, ANPR, billing
DATABASE	MONGO DB	Stores User's Plates, Sessions and Payments



PLATE RECOGNISER API

Detects license plates from uploaded frames

RAZORPAY API

Handles payments and verifications

FLASK ROUTES

Handle signup, login, ANPR, status, bill, pay

API'S AND **INTEGRATIONS**

PROTOTYPE

P

ParkSmart

Price Summary

₹1.24

Using as +91 99999 99999

>

Secured by

Razorpay

Payment Options

...

×

Recommended

UPI

Cards

EMI

Netbanking

Wallet

Pay Later

More options

UPI QR

11:38

Scan the QR using any UPI App

By proceeding, I agree to Razorpay

Time	Image	License Plates
Oct. 28, 2025, 7:01 p.m.		KL07AH9981 99.60%
Oct. 28, 2025, 7:01 p.m.		KL07AH9981 99.90%
Oct. 28, 2025, 6:53 p.m.		RJ14CV0002 99.90%
Oct. 28, 2025, 6:53 p.m.		RJ14CV0002 99.90%
Oct. 28, 2025, 6:52 p.m.		HR26DQ5551 97.70%
Oct. 12, 2025, 5:26 p.m.		HR26DQ5551 99.40%

My Plates

Add and manage your vehicle license plates for faster checkouts.

Plate Number

e.g., XYZ 123

Add Plate

Added Plates

HR26DQ5551

RJ14CV0002

Parking Summary

Start Time

15:44:33

End Time

15:45:35

Duration

0:01:02

Total Bill

Rs 1.24

Pay Now

CHALLENGES

SOLUTION

Challenges	Solutions
Vehicle entry and exit automation	Used ANPR (Automatic Number Plate Recognition) with Plate Recognizer API for automatic gate control
Detection accuracy	Improved using high-resolution frames and API confidence threshold
Session tracking	Verified active parking session before assigning a new slot
Payment safety	Implemented Razorpay signature verification and backend validation
Live UI updates	Used JavaScript periodic fetch with DOM updates for live slot and payment status

WHY USERS WILL ADOPT IT

- Saves time – no manual check in/out.
- Fair billing – pay only for exact duration.
- Easy for anyone to use.

SOCIAL VALUE

- Brings honesty, fairness, and transparency to daily financial transactions.
- Reduces disputes and frustration caused by inconsistent human billing.
- Encourages digital literacy and cashless payment adoption.
- Aligns with Digital India and Smart Governance initiatives.

SCOPE

- India has over 80 million private vehicles (growing rapidly).
- Urban cities face daily parking congestion.
- Target market: residential societies, malls, offices, and public parking lots.
- With digital integration, this can expand to smart city infrastructure.



OUTCOME

A CONTACTLESS SMART PARKING SYSTEM
WITH AUTOMATED ENTRY/EXIT DETECTION,
REAL-TIME BILLING, AND SECURE DIGITAL
PAYMENTS.

THANK YOU



PARKSMART

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Bedangi :- <https://github.com/bedangi>

Arnav :- www.linkedin.com/in/arnav-jain-767207274